Landau theoretical minimum: Paper 003

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The Apartment 50 preface is fictional staging. These eight teaching levels are an original curriculum, not a historical Landau examination. Use the study guide before attempting the problems; keep the separate instructor key closed.

For every answer, identify the declared input, calculation or counterexample, and the scope of the conclusion. Use integers and rational fractions for exact work. Decimal approximations are permitted only for the confidence-bound interpretation in Q23. Missing verification is an explicit open obligation.

Level 0: Know what has been decided

Q01. A phrase detector flags a proposal; a policy holds it; a route checker later accepts its witness. Explain the three decisions and identify whether there was false suppression if the hold permanently blocked submission.

Q02. The AM-FSP-2 list includes `particle-hole symmetry on triangle`. Compare its findings on "There is particle-hole symmetry on triangle plaquettes" and "There is no particle-hole symmetry on triangle plaquettes." What can the phrase comparison establish about the truth of either sentence?

Q03. Declare a directed-graph carrier with vertices `a,b`. Route I requires irreflexivity; route L permits loops. The same proposal adds edge `(a,a)`. Give its defect and valid near-miss status under these two contracts. List what a loop-refusal detector must bind before admission, and explain why a near miss from route L alone cannot qualify the detector for route I.

Level 1: Compute the policy you actually have

Q04. Reconstruct the baseline/Appendix A gate for all six cases: no signals; Signal 1 alone; Signal 2 alone with LOW/MEDIUM severity; Signal 2 alone with HIGH/CRITICAL severity; both with LOW/MEDIUM; both with HIGH/CRITICAL. Give the exact decision label and whether it refuses.

Q05. Compare that gate with `code/danger_gate.py`. Evaluate (a) one pattern match plus a MEDIUM Signal 2 record of defect count zero, and (b) one pattern match plus a MEDIUM record of defect count one. Identify both implementation differences. Is either table the strict Boolean conjunction of two signals?

Q06. Let `m` be the match count and set Signal 2 to absent when `m=0`, otherwise to `("HIGH",m)`. Compute the reachable decisions. Then change the derivation so Signal 2 is absent for every `m`. What changes? Explain why separately named inputs or four reachable test outcomes do not prove independent derivation in a deployed pipeline.

Level 2: Trace the host without inventing acceptance

Q07. At the canonical baseline, start with fuel 12, phase INTAKE, empty coverage, and no witness. Submit one batch: cover `elem_all`; propose `vertex_bits=[0,1,0]`; VERIFY. Give the decision, final semantic state, remaining fuel, and whether the witness has been proved wrong.

Q08. Start again and append COVER_ELEMENT after VERIFY. Give the decision and fuel. Compare an empty batch and the moment of exhaustion at the canonical baseline with the package host. State which implementation each result belongs to.

Q09. After reaching TERMINAL, the baseline host still records a receipt for each further attempted batch. Does fuel 12 bound all receipt count, retained bytes, or wall-clock time? State the bound fuel does establish and the additional limits a caller would need.

Level 3: Read evidence with its source attached

Q10. An editor changes a package receipt's fuel field and recomputes its unkeyed digest chain. Explain why the package verifier may accept it, what the canonical verifier additionally checks, and what neither verifier authenticates. Distinguish a digest edit without recomputation from a fully recomputed receipt.

Q11. A stored historical pilot reports terminal certificates, while the audited host returns NO_NATIVE_PORT. May the two records be pooled into an effectiveness estimate? Specify what would be needed to reproduce the historical certificates and to claim a current randomized treatment effect.

Q12. Prepare a source table for the deposit reviewed at `e11c07ce90318309f058c8097ffd1c440a8ffb84`, the canonical host baseline `52cb5457f6c485ba548cfd292f6012a941ed22cf`, the Appendix A audit `30609de2c3928238715375a9bac608256fa381d8`, and the historical correction `fdc0035a81c45e10b900fc917b8b0710df81391f`. Does any of those pins alone establish the behavior of today's default branch? Explain what a fresh test receipt must add.

Level 4: A sum is not a count of witnesses

Q13. For `n=2,K=2,t=1/2`, compute the Hill ratio for signal lists `[1,1/2,-7]` and `[2]`. Show the exact activation threshold and explain why negative entries do not cancel positive entries in this definition.

Q14. Keep `K=2,t=1/2` and cap each positive signal at `3/4`. Find the least possible number of contributing entries. Does that prove as many independent agents agreed? Also assess two unit entries copied from one source.

Q15. Derive the threshold for `n>0,K>0,0<t<1` by clearing the denominator. Handle `t=0` and `t=1` separately. Would `n=True` or `n=3/2` be a legal exponent

for the package's exact integer-exponent helper, even though the mathematical lemma permits positive real exponents?

Level 5: Screen the predicate that will run

Q16. Let `U={a,b,c}`, screened predicate `P={b}`, executed predicate `E={a,b}`, and valid self panel `S={a}`. Is the detector admitted by the screen? Does it fire on the panel? Supply the smallest counterexample.

Q17. Prove that screening is sound for every finite self panel if and only if `E(x) implies P(x)` for every input. Use a singleton panel for necessity. Is equality required for this one property?

Q18. Derive the additional implication needed to avoid rejecting a rule that would not fire on a panel. Then let `E=P={b}` and `S={a}`: if a new independent checker establishes that `b` is valid, what has admission failed to guarantee? Distinguish predicate agreement from panel coverage.

Level 6: Repair the physics claim's scope

Q19. For the manuscript's matrix `H=[[0,1,-1],[1,0,1],[-1,1,0]]`, compute `SHS` with `S=diag(1,-1,1)` and the cycle product of `sgn(-H_ij)`. Repeat the cycle-product test when all three off-diagonal bonds are positive. What separates the two triangles? Why does the standard local particle-hole sign assignment still fail on an odd cycle?

Q20. For the attractive onsite Hubbard model with real, spin-independent hopping, equal spin chemical potentials, and a real charge-channel decoupling, explain why equal spin-sector determinants give nonnegative configuration weights on a non-bipartite lattice. Does this establish the local particle-hole symmetry of Q19? What assumption is missing from "equal populations alone imply sign freedom"?

Q21. Classify three claims: a restricted model has a checked sign-free formulation; every instance in the NP-hard sign-problem class has a generic polynomial-time exact solution without complexity consequences; changing to a Majorana basis alone ensures positive weights. State the missing obligations, including why the hardness result is conditional on the complexity assumptions and why a phrase match cannot verify the Majorana or meron conditions.

Level 7: Design a study that can answer its question

Q22. Ten allocated runs yield three verified witnesses, two refusals, one timeout, and four unavailable/unknown outcomes. Compute the primary success fraction. Separately, of 20 independently valid eligible proposals, two were blocked; 100 invalid proposals were also blocked. Compute false suppression and explain why the invalid proposals cannot enlarge its denominator.

Q23. With zero false suppressions in `n` independent opportunities, derive the one-sided exact upper 95% bound. Evaluate `n=8`. Show using `(99/100)^n` why 298 opportunities miss the strict 1% gate and 299 meet it. Can 1,000 tagged rows generated from 40 base strings be substituted for 1,000 independent opportunities? Does a separate 95% bound on each of three routes automatically give a simultaneous 95% guarantee?

Q24. Draft the smallest confirmatory protocol that identifies the D-minus-R contrast. Include allocation/dependence unit, disjoint discovery and held-out instances, frozen source/detector/checker versions, common budgets and feedback opportunity, shadow checks, success and safety gates, missing outcomes, and stopping rules. Explain the ceiling if the route adapters remain unavailable or the assisted arm A supplies solutions.

Grading and advancement

Score each question from 0 to 4: 0, absent or wrong; 1, a label without support; 2, a partly correct calculation or argument; 3, correct result with its contract and evidence limit; 4, complete reasoning plus the requested counterexample or source binding. Maximum: 96.

Mastery requires at least 3 on every question and correction of every unsupported promotion of a finding, receipt, refusal, finite panel, or historical run. A high total cannot compensate for claiming a certificate from NO_NATIVE_PORT. This rubric evaluates the declared exercises; it grants no research or operational authority.